

Harshitha Vutukuru Muralikrishna

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PROFESSIONAL SUMMARY

Data Science and Machine Learning professional with 3+ years of experience in building ML models, ETL pipelines, and cloud-based deployments. Skilled in Python, TensorFlow, PyTorch, NLP, AWS SageMaker, and MLOps workflows. Proven ability to deliver data-driven solutions that improve accuracy, reduce latency, and enhance business outcomes.

EDUCATION

California State University, Los Angeles

Jan 2023 - Dec 2024

Master of Science in Computer Science

- Relevant Coursework: Advanced Algorithms, Data Science, Artificial Intelligence, Machine Learning, Advanced Web Programming, Operating Systems, Advanced Computer Networks

RNS Institute of Technology

Jul 2015 - Jun 2019

Bachelor of Engineering in Information Science and Engineering

- Relevant Coursework: Algorithms, Operating Systems, Data Science, Artificial Intelligence, Computer Networks

TECHNICAL SKILLS

- Programming Languages: Python, Java, JavaScript, SQL, HTML, CSS
- Machine Learning & Deep Learning: TensorFlow, PyTorch, Scikit-learn, Keras, PySpark
- NLP & LLMs: Hugging Face Transformers, LangChain, OpenAI APIs, FAISS
- Cloud & MLOps: AWS SageMaker, AWS Lambda, Kubernetes, Docker, Jenkins, CI/CD Pipelines
- Databases: MySQL, SQLite, PostgreSQL
- Frameworks & Tools: Flask, Spring Boot, RESTful APIs, Git, Tomcat

PROFESSIONAL EXPERIENCE

Accenture - Application Development Analyst | Bengaluru, India | Sep 2019 - May 2022

- Built and optimized SQL-based ETL pipelines for large-scale datasets (>10M records), improving backend performance and ML model training efficiency.
- Integrated predictive fraud detection models into Java/Spring Boot APIs, enabling real-time scoring in financial transactions.
- Developed Python/scikit-learn proof-of-concepts for anomaly detection in transaction data, improving fraud detection accuracy in pilot tests.
- Applied Test-Driven Development to enhance data quality and feature engineering workflows, reducing processing errors by 20%.

Spectrus Informatics - Full Stack Intern | Bengaluru, India | Feb 2018 - Apr 2018

- Developed a data-enabled e-commerce platform using Angular 2, JavaScript, and SQL, reducing page load times and optimizing user transaction workflows.
- Designed and implemented SQL databases and CRUD APIs to efficiently store and retrieve product, customer, and transaction data — laying groundwork for future analytics/ML integration.

PROJECTS

Brain Tumor Detection using CNNs

- Designed and trained deep learning models (ResNet50, VGG19, HALNet variants) for MRI-based tumor classification, achieving 95% accuracy.
- Applied preprocessing (augmentation, normalization, edge detection) to improve generalization and reduce false positives.
- Evaluated using ROC-AUC, Precision, Recall, and F1-score for diagnostic reliability.
- Skills/Tools: Python, TensorFlow, Keras, OpenCV, NumPy, Pandas, Matplotlib, Scikit-learn
- GitHub Link - https://github.com/harshithavutukuru/Datascience_project/tree/main

Retrieval-Augmented Generation (RAG) Pipeline

- Integrated OpenAI GPT-4 with FAISS and LangChain for domain-specific knowledge retrieval.
- Used Pinecone as vector database for faster and more accurate context retrieval.
- Measured system performance with BLEU, ROUGE, and retrieval precision metrics.
- Skills/Tools: Python, LangChain, FAISS, Pinecone, OpenAI API, Hugging Face Transformers, NLP, REST APIs
- GitHub Link - <https://github.com/harshithavutukuru/RAG-based-chatbot>

AWS SageMaker ML Deployment

- Developed end-to-end ML pipeline with AWS SageMaker, Lambda, and S3 for automated model training and deployment.
- Deployed fine-tuned GPT-3 chatbot on AWS with scalable architecture.
- Automated hyperparameter tuning using SageMaker's optimization tools.
- Skills/Tools: AWS SageMaker, AWS Lambda, AWS S3, Python, Boto3, Docker, Git, MLOps
- GitHub Link - <https://github.com/harshithavutukuru/AWS-SageMaker>

CERTIFICATIONS

AWS Certified Machine Learning – Specialty (In Progress)